

Muhammad Farooqi

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EDUCATION

University of Waterloo

Mechatronics and Robotics Engineering + AI Option — Grade: **91%**

Relevant Courses: OOP in C/C++, Data Structures and Algorithms, Digital Logic, Linear Circuits, Linear Algebra.

Waterloo, ON

Expected Apr-2028

EXPERIENCE

Robot Engineer Lead

Feb 2025 – Present

WATOnomous

Waterloo, ON

- Developed a **Quest2** teleop pipeline that streamed **ROS2** commands to sim/real robots via rosbridge in **Isaac Sim**.
- Collected **120** demos and fine-tuned an **ACT** policy in **LeRobot** for a letter-drawing task on an SO-100 arm.
- Converted a custom arm from **URDF** to **USD** and authored a drawer-opening task environment in **Isaac Lab**.
- Trained the custom arm with **PPO** in **Isaac Lab** for drawer-opening manipulation in simulation.
- Implemented IK solver in **Isaac Lab** with **MuJoCo** solver, converging a 21-DOF hand to **0.1mm** tolerance.
- Optimized depth estimation with **ONNX** and **TensorRT**, reducing latency **29%** and increasing from **85** to **120 FPS**.

Navigation Software Engineer Intern

Sep 2025 – Apr 2026

Rivian

Vancouver, BC

- Architected a shift-left testing pipeline introducing automated smoke tests on PRs, replacing 3-hour **HIL** image builds with 30-min APK builds, spinning up **45 CI jobs** daily and preventing regressions in **30%** of PRs before merge.
- Extended the **APK** pipeline with **AI agents** and a custom ADB MCP server to autonomously run targeted **HIL** tests across **R1** and **R2** platforms, replacing manual integration testing and cutting daily testing time by **50%**, driving adoption across **6** teams org-wide and presented to the **CTO** and multiple **directors**.
- Owned test planning for the Driver Display swap feature, authoring **100** cases validating safety-critical **ADAS** feature visibility across view transitions, button sequences, and map state changes, ensuring correct system behavior pre-release.
- Increased automated navigation test coverage by **117%** in nightly **CI** runs, covering reroutes, destination edits, and display changes, building regression coverage from scratch and saving **~5 hrs** of manual testing per release.

Robotics Software Engineer Intern

May 2025 - Aug 2025

Lincoln Electric Automation

Waterloo, ON

- Designed and deployed an end-to-end tack welding program for 6-axis **Fanuc** (KAREL) and **ABB** (RAPID) arms, removing separate welder setup and saving **5–7 seconds** per tack, **2–3 minutes** per part across **2** production cells.
- Developed motion control and weld parameter interfaces in **TypeScript** using **functional programming**, integrating with **Controller APIs** for 3 tack sizes and validating through **100+** virtual welds in **RobotStudio** and **RoboGuide**.

Backend Software Developer Intern

Sep 2024 – Dec 2024

ISARA

Waterloo, ON

- Built cryptographic risk analysis algorithms for TLS, SMB, and plaintext protocols using **Node.js** and **GraphiQL**.
- Automated demo deployment using a **Python** script to fetch build from an **AWS S3** bucket and developed an admin control panel with **React**, **PostgreSQL**, and **Docker**, saving **10 minutes** per deployment.

Computer Vision Intern

Jan 2024 – Apr 2024

AirMatrix

Mississauga, ON

- Annotated over **3,000 traffic images** using **Roboflow**, achieving **95%** accuracy at **0.5 IoU** on vehicle detection.
- Trained a **YOLOv8 vehicle detection model** in **Python** for vehicle detection, counting, and speed estimation.

PROJECTS

SLAM ROS2 Robot | *Linux, ROS2, Python, Gazebo, Rviz, Camera/Lidar, Raspberry Pi, Arduino*

- Created **URDF** files with collision and inertia values for realistic simulation in **ROS2** and **Gazebo**.
- Configured a differential drive robot in **Rviz** with **camera** and **2D LiDAR** nodes, generating point clouds from LiDAR and applying **multi-sensor** fusion with **SLAM Toolbox** for mapping and localization.
- Enabled teleoperation with a **PS4 controller** via the **joy** topic using the **ros2 control** library.



TECHNICAL SKILLS

Programming Languages: Python, C++, Rust, TypeScript, JavaScript, SQL, HTML/CSS

AI & ML: PyTorch, OpenCV, NumPy, ONNX, TensorRT, YOLOv8/v11, HuggingFace, MediaPipe, nuScenes

Robotics & Simulation: ROS2, Isaac Sim, Foxglove, Rviz, Gazebo, Isaac Lab, Mujoco, SLAM, Arduino, Raspberry Pi

Developer Tools: Git, Docker, Linux, CUDA, Jupyter Notebook, Claude, Cursor, MCP, Figma, Ollama